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EXTENDED ABSTRACT

Projected changes in the Arctic sea-ice seasonal cycle inferred from clustering

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1. Introduction and objectives

The Arctic has been identified as the region on Earth warming the fastest since the 1980s [Rantanen et al. (2022)]. The Arctic sea ice melt caused by this warming also contributes to its intensification, as the open ocean expands and absorbs more solar radiation than sea ice. This is known as the surface albedo feedback [Goose et al. (2018)]. These rapid and significant changes, primarily driven by anthropogenic greenhouse gas emissions [IPCC (2021)], may have multiple consequences in the near future on ecosystems but also on the global climate [Cvijanovic et al. (2025)].

Hence, it appears critical to have a better understanding of the Arctic sea ice dynamics in order to anticipate how the region might be reshaped in the next decades. Various methods have been used to characterize the Arctic regions, including the trend in sea ice extent, the sea ice age, or the duration of the ice-free period. Recently, a new statistical approach has been proposed by [Simon et al. (2025)] to objectively regionalize changes in Arctic sea ice based on their seasonal cycles. By implementing an unsupervised k-means clustering of sea ice concentration (SIC) satellite measurements over the 1979–2023 period, four clusters corresponding to four typical sea-ice seasonal cycles were retrieved and their spatio-temporal distribution provided a unique way to describe the Arctic evolution.

In this study, we investigate how this regionalization of the Arctic could further evolve by the end of the century by applying this data-driven methodology to climate projections of the IPSL-CM6A-LR climate model [Boucher et al. (2020)]. We assess the clusters and their future spatio-temporal evolution in a historical simulation (1850–2015) followed by climate projection driven by three different scenarios (2015–2100).

2. Methods and data

We use outputs from the IPSL-CM6A-LR climate model, providing monthly mean SIC at a 50-km resolution. We consider the "historical" simulation – corresponding to the 1850-2015 period – and three distinct future socio-economic scenario pathways "ssp126", "ssp245" and "ssp585" simulating an anthropogenic radiative forcing of 2.6 W.m^{-2}, 4.5 W.m^{-2} and 8.5 W.m^{-2} by 2100, respectively. The studied region covers the ocean north of $55°$N.

We build a common training dataset with the historical dataset for the 1850-2015 period, and by selecting one out of every three years from each scenarios for the 2015-2100 period. This ensures that each scenario contributes equally to the optimization of the centroids coordinates.

This k-means clustering is performed on the SIC seasonal cycle at each grid cell updated each month. The algorithm relies on the Mahalanobis distance to constrain the clustering with prior information on

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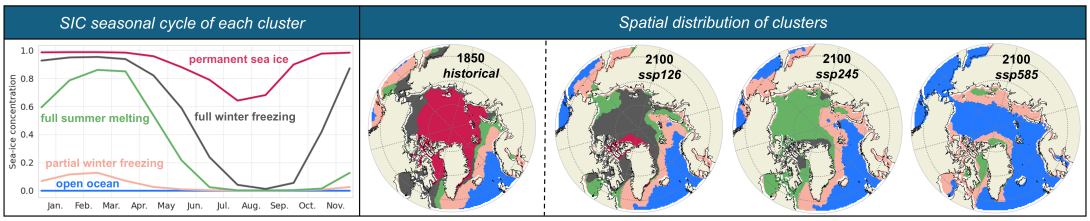

Figure 1. *Left panel: SIC seasonal cycle for each cluster. Right panel: spatial distribution of clusters in 1850 and in 2100 for each scenario.*

the SIC seasonal cycle and its initialization uses equal quantile separation [Simon et al. (2025)]. We evaluate the results for different number of clusters by using a Silhouette coefficient.

3. Preliminary results

Based on the Silhouette coefficient, we choose 5 clusters to best represent the Arctic evolution over the 1850–2100 period. They correspond to: open ocean, partial winter-freezing, full summer-melting, full winter-freezing, and permanent sea ice (Fig. 1). The clusters and their distribution indicate that the ice-free period tends to lengthen over time. Further, sea ice growth happens later in the year and becomes more and more partial. Some critical years defined by the disappearing of one or more clusters can be detected depending on the scenario.

Regardless of the scenario, our approach uses the same centroids. However, the spatio-temporal distribution of clusters are very different from one scenario to another. While the permanent sea ice cluster disappears by ∼2050 in every scenario, only the ssp126 scenario (i.e., the most sustainable scenario) is able to retain a constant distribution of the full winter-freezing and the full summer-melting clusters by 2100. In contrast, these clusters disappear successively with the ssp585 scenario (i.e., the most intense scenario). Under this pathway, the central Arctic becomes ice-free by the end of the century.

This uni-variate statistical analysis provides some coherent insights on the evolution of the Arctic sea ice cover for various climate scenarios. While it may already offer promising results for the climate research community, enriching it by adding other key variables (e.g., Sea Ice Thickness) will be the goal of future studies.

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